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SHALLOW MAGNETIC INCLINATIONS IN THE CRETACEOUS VALLE GROUP, BAJA CALIFORNIA: REMAGNETIZATION, COMPACTION, OR TERRANE TRANSLATION?

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Abstract. Paleomagnetic data from Albian to Turonian sedimentary rocks on Cedros Island, Mexico (28.2° N, 115.2° W), support the interpretation that Cretaceous rocks of western Baja California have moved farther northward than the 3° of latitude assignable to Neogene oblique rifting in the Gulf of California. Averaged Cretaceous paleomagnetic results from Cedros Island support 20 ± 10° of northward displacement and 14 ± 7° of clockwise rotation with respect to cratonic North America. Positive field stability tests from the Vizcaino terrane substantiate a mid-Cretaceous age for the high-temperature characteristic remanent magnetization in mid-Cretaceous strata. Therefore coincidence of characteristic magnetization directions and the expected Quaternary axial dipole direction is not due to post mid-Cretaceous remagnetization. A slump test performed on internally coherent, intrabasinal slump blocks within a paleontologically dated olistostrome demonstrates a mid-Cretaceous age of magnetization in the Valle Group. The in situ high-temperature natural remanent magnetization directions markedly diverge from the expected Quaternary axial dipole, indicating that the characteristic, high-temperature magnetization was acquired prior to intrabasinal slumping. Early acquisition of the characteristic magnetization is also supported by a regional attitude test involving three localities in coherent mid-Cretaceous Valle Group strata. Paleomagnetic inclinations in mudstone are not different from those in sandstone, indicating that burial compaction did not bias the results toward shallow inclinations in the Vizcaino terrane.

1. INTRODUCTION

Two recent papers by Gastil [1991] and Beck [1991] summarize an apparent paradox in the tectonic history of Baja California (Figure 1). Cretaceous and early Cenozoic paleomagnetic directions from both sedimentary and igneous rocks suggest that the region lay 10°-25° to the south with respect to stable North America (see summaries by Hagstrum and Filmer [1990] and Beck [1991]). On the other hand, Paleozoic roof pendants within the Cretaceous Peninsular Ranges batholith correlate well with rocks located nearby across the Gulf of California [Gastil, 1991]. The Paleozoic correlation diminishes the likelihood of large-scale translations suggested by the paleomagnetic data, unless we invoke multiple episodes of terrane motion and a serendipitous realignment of the Paleozoic strata before Neogene rifting in the Gulf of California [Gastil et al., 1986; Gastil, 1981, 1991].

Resolution of the paradox depends upon testing the robustness of the mutually incompatible geological and paleomagnetic data sets. Three viable hypotheses forwarded to explain the inclinations without invoking terrane motion are (1) undetected tilting of plutonic rocks [Butler et al., 1991], (2) recent remagnetization of paleomagnetic sites, and (3) compaction shallowing of sedimentary rocks [Butler et al., 1991]. Our work tests the second and third alternate hypotheses in Cretaceous Valle Group strata of the Vizcaino terrane.

2. REGIONAL GEOLOGIC SETTING

"Middle" Cretaceous outer neritic to bathyal marine forearc basin strata of the Vizcaino Peninsula and Cedros Island, Mexico, form an overlap sequence tying together Triassic to Jurassic paleo-oceanic basement terranes. The basement rocks and overlying strata (Valle Group) form the composite Vizcaino tectonostratigraphic terrane of Coney et al. [1980] (Figure 1), which was likely accreted to the North American margin in Late Jurassic time [Kimbrough, 1982, 1985; Barnes, 1984; Boles and Landis, 1984; Moore, 1985] (Figure 1). Uncertainties in the timing of accretion and the possibility of post accretion strike-slip motion along the edge of North America warrant caution in applying the paleomagnetic results of the Vizcaino terrane to other terranes of Baja California.

Valle Group strata are unconformably overlain by Miocene and Pliocene deposits on both the Vizcaino Peninsula and Cedros Island. On the Vizcaino Peninsula the Miocene deposits were used to limit the age of magnetization in three sites of the Valle Group to pre-Miocene time via a fold test [Hagstrum et al., 1985]. On Cedros Island the Miocene and Pliocene strata are moderately tilted, making a similar field stability test problematic.

3. PREVIOUS PALEOMAGNETIC RESULTS FROM VIZCAINO TERRANE

Patterson [1984b] and Hagstrum et al. [1985] provided the first paleomagnetic studies of the Valle Group suggesting low paleolatitudes for the Vizcaino terrane in Cretaceous time. Out of six sites studied in the initial work [Patterson, 1984a, b], in situ directions for only two sites are published. For one of those two sites, the only previous result from Cedros Island, the in situ direction is indistinguishable from the average Quaternary direction. Thus the only evidence provided for a Cretaceous age of magnetization is between-site consistency in the structurally corrected directions. Hagstrum et al. [1985] found 12 of the 15 sites they sampled in the Valle Group on the Vizcaino Peninsula to have been remagnetized by postdepositional mineralization. Magnetization of the remaining three sites is considered to predate deposition of overlying Miocene strata based on a

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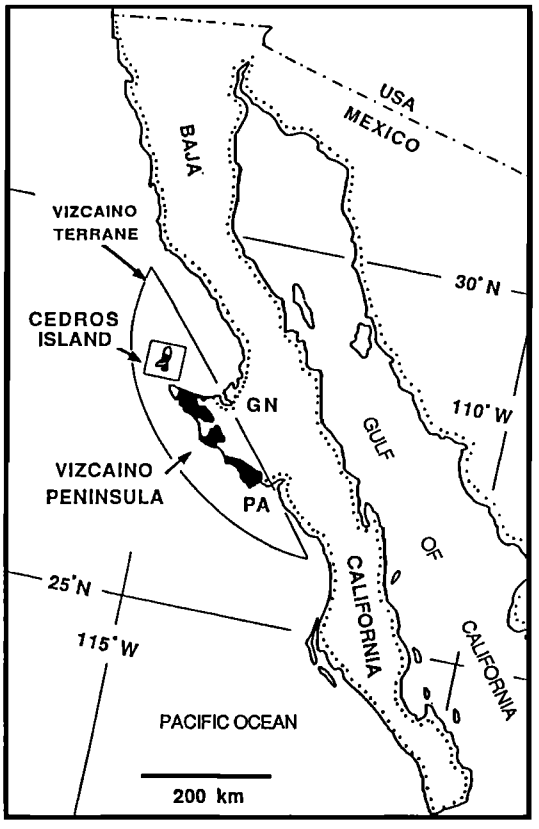


Fig. 1. Location of Cedros Island in the borderland province of Baja California. The Valle Group of the Vizcaino terrane is shown as solid areas. GN, Guerrero Negro; PA, Punta Abreojos.

positive fold test and similarity to Patterson's [1984b] results. The high proportion of remagnetized Valle Group sites reported by Hagstrum et al. [1985] cautions against blind acceptance of the four out of six sites for which Patterson [1984a, b] does not include in situ results. Our work lends further credence to the previous studies by providing additional paleomagnetic sites showing similar results and by adding field stability tests to the collective data set. These tests strongly suggest a middle Cretaceous age of magnetization in middle Cretaceous rocks of the Vizcaino terrane.

4. PALEOMAGNETIC SAMPLE LOCALITIES ON CEDROS ISLAND

Albian to Coniacian strata of the Valle Group crop out in three faulted and tilted structural blocks on Cedros Island (Figure 2) [Smith and Busby, 1993]. Paleomagnetic samples come from four localities spanning two of the structural blocks: Valle Group 1, Valle Group 2, Pinos Formation 1, and Pinos Formation Olistostrome 1 (VG-1, VG-2, P-1, and PO-1; Figure 2). The biostratigraphic ages of the localities are reported by Smith et al. [1993].

Sample localities VG-1 and VG-2 are from nonslumped shale and thin-bedded sandstone of the undivided Valle Group strata located along the southern intertidal zone of the southeastern outcrop belt (Figure 2). These two sample localities are between the ages of early Cenomanian to early middle Turonian, with the younger age more likely. Locality PO-1 is within an intrabasinal olistostrome in the Pinos Formation located along the northwestern edge of the northern outcrop belt (Figure 3). The olistostrome comprises large blocks of interbedded shale and thin-bedded sandstone that range in age from Albian to middle Cenomanian. The strata overlying the olistostrome bear nannofossils that range in age from early Cenomanian to early Coniacian. Thus the slumping event that produced the olistostrome is limited to post mid-Cenomanian (youngest known block) but is pre-early Coniacian in age. The slump blocks have smeared

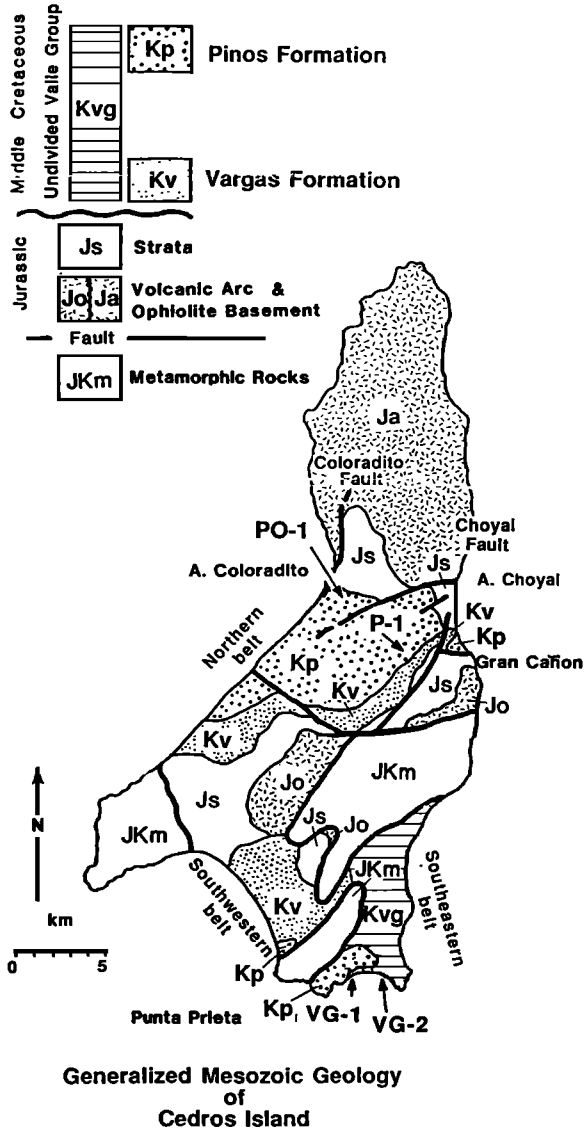


Fig. 2. Generalized geologic map of Cedros Island (modified from Kilmer [1984]), showing paleomagnetic sample localities VG-1, VG-2, PO-1, and P-1. See text for discussion of localities

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margins, and some are severely folded, supporting the above biostratigraphic indication that slumping occurred shortly after initial deposition of the strata.

We include recalculated data from Patterson's [1984a,b] Cedros Island paleomagnetic site in the present study (P-1 in Figure 2). Although Patterson [1984a,b] did not precisely plot the location of P-1, we offer the following description based on Patterson's [1984a] and our familiarity with the region. Sample locality P-1 is within the lowest 45 m of the Pinos Formation. This section, located inland from the mouth of Gran Cañon (Figure 2), comprises medium- to thick-bedded sandstone with minor interbedded mudstone and slumped shale. The strata are early Cenomanian in age [Smith et al., 1993] rather than Turonian, as reported by Patterson [1984a,b].

5. PALEOMAGNETIC STUDY

5.1. *Field and Laboratory Techniques*

Oriented hand samples of Valle Group strata were collected for paleomagnetic analysis from three localities on Cedros Island (Figure 2). Fresh-looking samples were selected to avoid surface alteration. Our collection was limited to well-lithified rocks that ranged in grain size from silty laminated mudstone to medium-grained sandstone. Cores were drilled in the laboratory, and specimens were then made by cutting cores into several cylindrical pieces. Most specimens were siltstone or very fine grained sandstone, and many comprised sandstone-mudstone couplets.

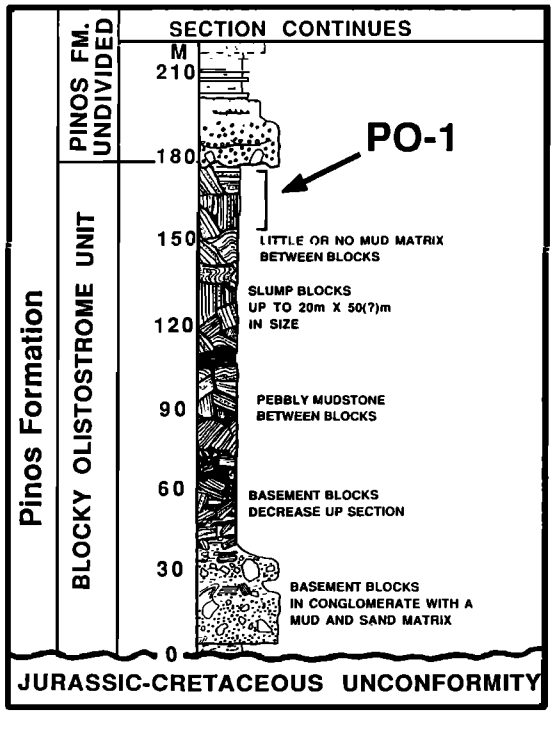


Fig. 3. Stratigraphic position sample locality PO-1 in olistostrome of Pinos Formation (see Figure 2 for location).

All specimens were treated with progressive thermal demagnetization. Measurements were made using a cryogenic magnetometer (2G Enterprises model 760-R) located in a magnetically shielded room at the University of California, Santa Barbara. Pilot samples from each bed (usually from each hand sample) were demagnetized, and the remanent magnetization was measured at 25°C intervals to determine the demagnetization behavior and optimum temperatures for processing the remaining specimens. Specimens were discounted from statistical analysis if they did not yield a well-defined, high-temperature magnetization. A stable high-temperature magnetization was inferred in specimens displaying a decay of remanence intensity concomitant with little change in magnetic direction.

5.2. *Magnetic Behavior*

From locality PO-1 we used 82 specimens from 43 hand samples of 24 beds representing six intraformational slump blocks. The samples from PO-1 span at least 21 m of composite stratigraphic thickness. We selected blocks with a wide range of structural orientations to be used in the slump test, including two that were stratigraphically overturned. Specimens from locality PO-1 typically have initial natural remanent magnetization (NRM) intensities of 5×10^{-3} A/m and in situ directions similar to the expected average Quaternary direction for Cedros Island. The initial NRM direction is a combination of two components: a viscous(?) remanent magnetization (VRM) removed between 250°C and 550°C and a characteristic high-temperature component with a direction that persists until about 610°C (Figure 4a). All specimens but one exhibited a systematic decrease in magnetic intensity with progressively higher temperatures (Figure 4d).

From localities VG-1 and VG-2 we used a total of 30 specimens from 10 hand samples taken from 10 beds. We estimate that our sampling spans about 10 meters of stratigraphic thickness at each of the two localities, but the stratigraphic separation of the sites is unknown, owing to the presence of innumerable high- and low-angle normal faults along the southern coast of Cedros Island. Specimens from localities VG-1 and VG-2 have typical initial NRM intensities of 2×10^{-2} A/m and in situ directions distinct from the expected Quaternary direction. Difference vectors calculated at low temperatures, however, commonly show that the expected Quaternary direction is a magnetically soft component of the initial NRM. Thus the initial NRM includes a VRM(?) which is removed by about 250°C and a characteristic high-temperature component that remains stable until near 600°C (Figures 4b,c,d).

6. PALEOMAGNETIC RESULTS

The average characteristic remanence direction for each bed is used as an independent estimate of the mid-Cretaceous geomagnetic direction for Cedros Island. Because our averaged results include samples from 42 beds (Table 1) spanning Early Cenomanian to Turonian ages, we believe secular variation is averaged out of the overall mean paleomagnetic direction for Cedros Island. The following discussion refers to the characteristic high-temperature magnetizations unless otherwise noted.

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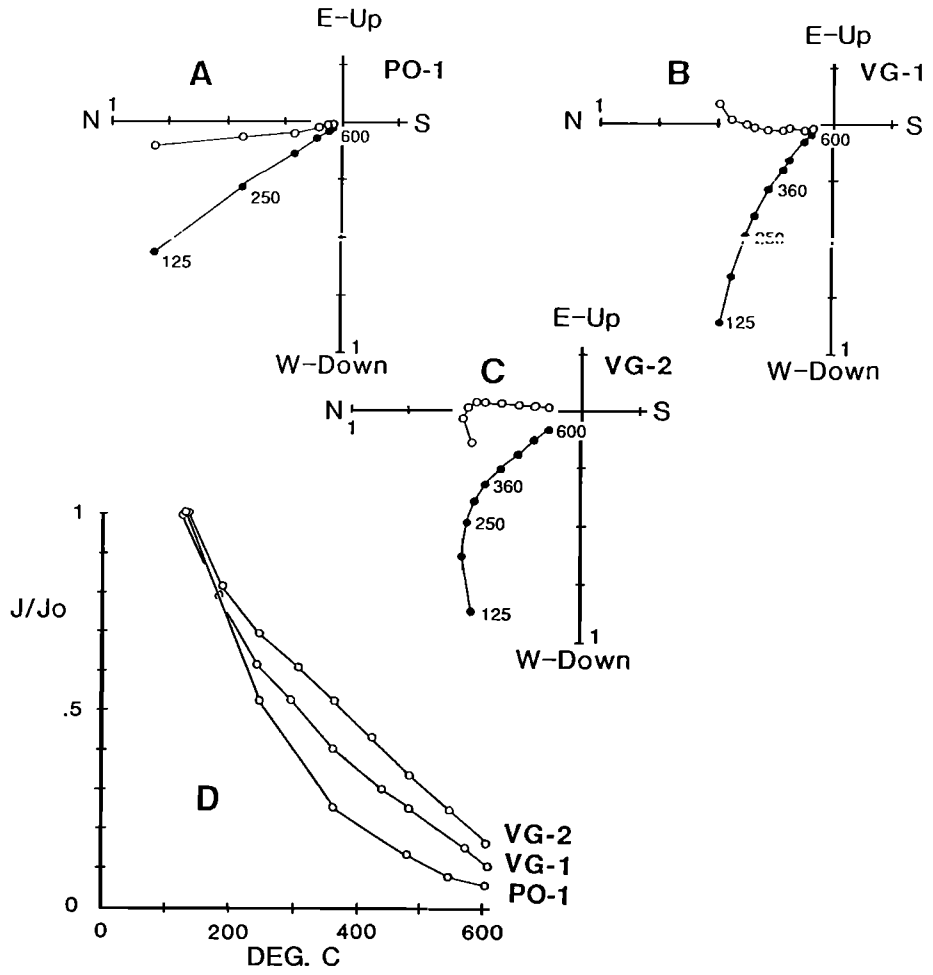


Fig. 4. Orthogonal vector diagrams for structurally corrected results and corresponding relative intensity curves (J/Jo versus temperature) illustrate simple thermal demagnetization behavior. (a)-(c) Orthogonal vector component diagrams; open circles represent the projection of the vector tip in map view (showing declination), and solid circles represent projection on the north-south vertical plane (showing projection of inclination). Each division on the axes represents 0.25 of the initial natural remanent magnetization intensity for each specimen; demagnetization level is in degrees Celcius. (d) Relative intensity plot shows the proportional loss of magnetic intensity J/Jo at progressively higher temperatures during continuous thermal demagnetization. J/Jo data correspond to the points in the vector diagrams.

6.1. *Regional Tilt Test*

Samples from localities VG-1 and VG-2 have in situ directions that differ from the expected Quaternary field direction (Figure 5). The precision of the mean at both localities is improved when structurally corrected (Table 1), and average directions from the two localities converge with the direction from locality P-1 (Figure 5). The convergence of directions constitutes a positive regional tilt test. This positive tilt test suggests that magnetization acquisition predates differential tilting of the northern and southeastern outcrop belts (Figure 2). Unfortunately, the age of tilting can only be limited to pre-Pleistocene time because marine terraces on the island are not significantly tilted. The three average directions (P-1, VG-1, and VG-2) are distributed on a small circle arc in stereographic projection (Figure 5). This dispersion likely reflects minor differential vertical axis rotations but does not greatly affect the outcome of this study.

6.2. *Slump Test*

Slump blocks in the olistostrome of locality PO-1 occur in a wide range of structural attitudes (Table 1). Sedimentary facing indicators within the blocks show that many are overturned. We attempted to sample from internally coherent intrabasinal blocks that would have resisted disaggregation and ductile deformation during slope failure and transport.

Figure 6a shows the structurally uncorrected initial NRM and high-temperature magnetizations for six slump blocks from the olistostrome. Initial NRM directions (x's in Figure

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TABLE 1. Paleomagnetic Data From Valle Group, Cedros Island

		In Situ					Corrected				
Site(Bed)	N	I,deg	D,deg	a95,deg	k	tilt	I,deg	D,deg	a95,deg	k	R,deg
(B1-1)	6	65	273	..	..	..	46	54	11	24	..
(B1-2)	6	63	290	..	..	..	43	58	6	68	..
(B1-3)	2	73	320	..	..	..	29	59	14	204	..
(B1-4)	2	82	245	..	..	..	28	72	25	66	..
(B1-5)	1	72	271	..	..	..	35	57	..	..	..
(B1-6)	2	71	245	..	..	..	39	66	17	145	..
Block 1	6	73	277	9	58	340;66E	40	61	8	79	..
(B2-1)	2	17	335	..	..	..	0	333	82	8	..
(B2-2)	2	37	352	..	..	..	24	349	30	48	..
(B2-3)	2	54	303	..	..	..	40	310	14	222	..
(B2-4)	3	59	293	..	..	..	46	304	14	54	..
Block 2	4	44	326	32	9	240;15S	29	262	32	9	..
(B3-1)	4	24	146	..	..	..	32	272	14	26	..
(B3-2)	6	24	163	..	..	..	30	296	15	13	..
Block 3	2	24	155	34	57	042;240E	32	284	48	30	..
(B4-1)	2	25	288	..	..	..	65	218	18	17	..
(B4-2)	2	16	297	..	..	..	63	248	16	33	..
(B4-3)	6	23	305	..	..	..	72	246	14	17	..
Block 4	3	22	297	14	82	050;65S	67	237	13	93	..
(B5-1)	4	38	64	..	..	..	27	358	6	131	..
(B5-2)	4	29	66	..	..	..	32	7	4	281	..
(B5-3)	2	47	84	..	..	..	31	348	20	104	..
(B5-4)	4	34	79	..	..	..	38	7	10	61	..
Block 5	4	37	73	12	58	217;80W	32	0	10	83	..
(B8-1)	3	27	50	..	..	..	5	348	18	32	..
(B8-2)	4	28	50	..	..	..	4	346	14	30	..
(B8-3)	5	14	64	..	..	..	17	7	11	31	..
(B8-4)	4	30	47	..	..	..	1	349	14	27	..
(B8-5)	4	28	58	..	..	..	10	345	13	32	..
Block 8	5	26	54	9	79	030;240E	7	351	10	57	..
PO-1 AVE	24	49	356.5	13	7	..	33		6	27	..
											..
(VG1-1)	1	-4	357	..	..	..	37	354	..	..	..
(VG1-2)	1	-20	353	..	..	..	43	351	..	..	..
(VG1-3)	3	-5	343	..	..	..	38	338	10	113	..
(VG1-4)	2	-27	7	..	..	..	30	6	16	152	..
(VG1-5)	2	-29	8	..	..	..	37	5	3	3536	..
(VG1-6)	1	-31	12	..	..	..	38	10	..	..	..
VG-1 AVE	6	-20	359	10	26	100;50S	37	357	6	62	14±7
(VG2-1)	3	-20	22	..	..	..	44	18	5	231	..
(VG2-2)	7	-35	5	..	..	..	30	6	4	157	..
(VG2-3)	8	-20	16	..	..	..	38	12	6	44	..
(VG2-4)	1	-17	13	..	..	..	40	7	..	..	..
VG-2 AVE	4	-20	15	11	43	120;40S	38	10	6	127	29±13
P-1 AVE	8	52	3	7	43	020;25W	39	348	11	35	6±12
AVE - 3	18	12	5	15	4	..	38	356	4	56	14±7

In situ and structurally corrected inclination (I) and declination (D) values for characteristic magnetizations of individual beds from localities PO-1, VG-1, and VG-2. N is the number of specimens per bed or the number of beds per site, respectively; a95 is radius of the 95% confidence circle [Fisher, 1953; Demarest, 1983]; k is precision parameter; tilt is the orientation of each site (overturned slump blocks have dip>90°); R is mean value and 95% confidence interval for clockwise rotation of site with respect to stable North America [Globerman and Irving, 1989] calculated using the methods of Beck [1989]; P-1 is recalculated from Patterson [1984a]. PO-1 AVE is based on inclination data only [McFadden and Reid, 1982]. AVE-3 is the average direction of VG1,VG2, and P1.

6a) cluster near the expected Quaternary field direction. This component of the NRM is likely an overprint acquired sometime after slumping. After thermal demagnetization, the average directions in individual blocks disperse (crosses in Figure 6a), indicating that the characteristic, high-temperature remanence magnetizations were acquired before slumping. That behavior constitutes a qualitative "conglomerate test," demonstrating that the overprint is removable. We do not recommend quantifying the conglomerate test in this case because the slump blocks are far from spherical in shape. Their shapes no doubt influenced their physical orientation in the olistostrome, precluding a meaningful test of randomness. As further evidence that the high-temperature directions are primary magnetic directions, we note that the structurally corrected directions of four of the blocks have strikingly similar mean inclinations (Figure 6b). The similarity is probably not coincidental, considering that the inclinations agree well with

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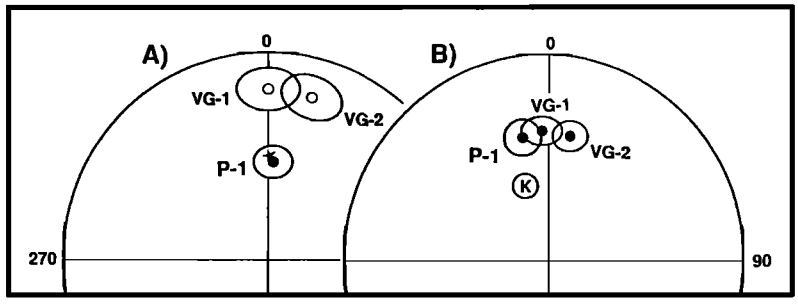


Fig. 5. Attitude test from nonslumped strata of Valle Group. Means and 95% confidence intervals for characteristic magnetic directions are from localities VG-1, VG-2, and P-1. Solid circles are projections from lower hemisphere, and open circles are projections from upper hemisphere. (a) In situ directions. The in situ direction for P-1 (recalculated from Patterson [1984a]) is indistinguishable from the Quaternary dipole direction (star). (b) Upon structural correction, the three results converge on a common direction, and precision increases (Table 1). K is the expected mid-Cretaceous magnetic field for North America.

inclinations from nonslumped sites of the Valle Group as well (3 in Figure 6b). The declinations measured in the slump blocks remain dispersed because of vertical axis rotations imparted during slumping.

6.3. *Influence of Burial Compaction on Magnetic Inclination in Vizcaino Terrane*

Shallow magnetic inclinations in the Valle Group on the Vizcaino terrane are interpreted to record a low paleolatitude for this region in middle Cretaceous time [Patterson, 1984b; Hagstrum et al., 1985]. This interpretation is conditional upon discounting other mechanisms for producing shallow magnetic inclinations in these strata. Butler et al. [1991] suggest that burial compaction of sediments has systematically biased the Valle Group data toward shallow inclinations. We test that hypothesis by looking for a relationship between magnetic inclination and sediment grain size in paleomagnetic samples of the Valle Group.

It is well known that finer-grained sediments compact much more than coarser-grained sediments do, given similar overburden conditions. Clay-sized sediment, for example, can lose nearly all initial porosity during compaction, which can exceed 80% by volume [Perrier and Quiblier, 1974]. Uncompacted, well-sorted sand, in contrast, has about 40% porosity, so the initial potential to lose volume is much less in sandstone. Moreover, the greater physical strength of siliciclastic sand grains and common presence of pore-filling cement (e.g., calcite) tend to prevent complete collapse of the sandstone framework during compaction. Therefore if magnetic inclination shallowing is driven by physical compaction, there should be a positive correlation between

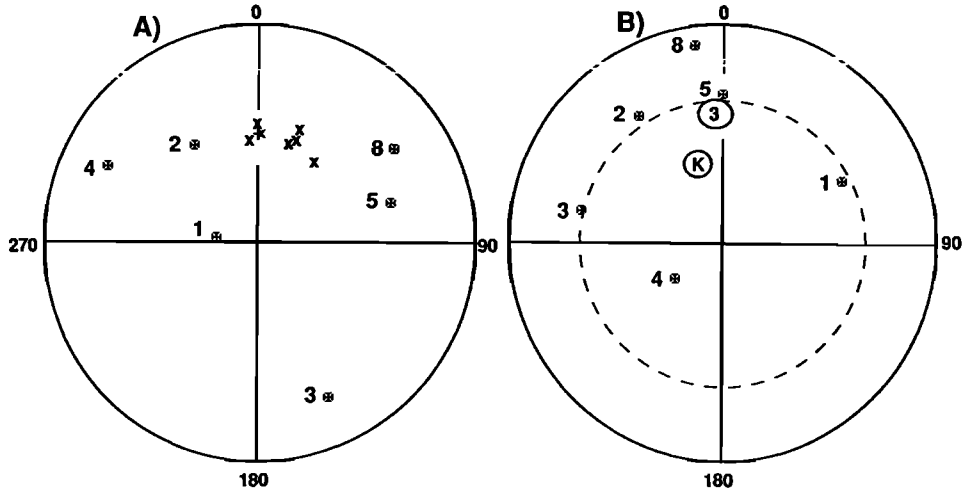


Fig. 6. Slump test in Pinos Formation olistostrome. Average directions are shown for each of six blocks (1, 2, 3, 4, 5, and 8) from the olistostrome at locality PO-1. (a) In situ initial NRM directions (x's) cluster near Quaternary dipole direction (star), but disperse upon thermal treatment (numbered crosses). (b) The inclinations of four blocks (1, 2, 3, and 5) converge upon an inclination of 33.3° (dashed small circle) after structural correction. The average inclination is flattened compared to the expected mid-Cretaceous North American pole (K) [Globerman and Irving, 1989]. The 3 is the mean direction and 95% confidence interval calculated from localities VG-1, VG-2, and P-1 (AVE-3 of Table 1).

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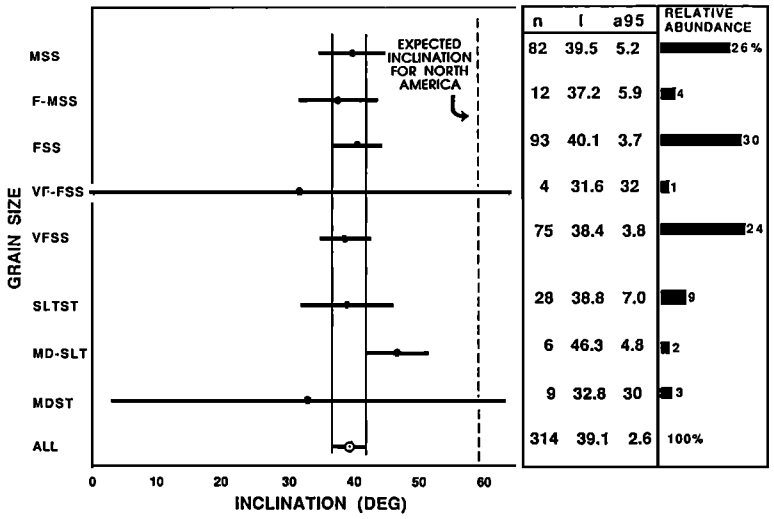


Fig. 7. Grain size versus inclination. Solid circles and horizontal bars are mean magnetic inclinations and 95% confidence limits for eight descriptive grain size classes. Vertical bars are 95% confidence limits on the grand mean of 314 samples (open circle): 281 samples are from Patterson [1984a], and 33 samples are representative of the present study. N, I, and a95 are the number of samples, average magnetic inclination, and 95% confidence limits for each grain size category calculated using the methods of McFadden and Reid [1982]. The relative abundance (out of 314 samples) of each size class is on the far right. Grain size classes are mudstone (MDST), silty mudstone (MD-SLT), siltstone (SLTST), very fine grained sandstone (VFSS), very fine- to fine grained sandstone (VF-FSS), fine sandstone (FSS), fine- to medium grained sandstone (F-MSS), and medium sandstone (MSS).

grain size and magnetic inclination in the rock samples from the Valle Group.

Magnetic inclination is plotted against grain size for 314 specimens from the Valle Group on both Cedros Island and the Vizcaino Peninsula (Figure 7). The data are presented as deviates from the appropriate site mean, with the grand mean of all samples added back in. This method preserves the effects of grain size by eliminating variation stemming from true between-site differences. The calculated confidence limits for average inclination values fail to show a significant difference between means ($P>0.05$) measured in samples grouped by grain size. We conclude that compaction alone is not likely to have greatly influenced magnetic inclination values in the Valle Group samples.

7. CONCLUSIONS

Strong similarities exist between Cretaceous paleopoles and the calculated modern axial dipole directions for most localities in Baja and southern California [Patterson, 1984b; Hagstrum et al., 1985; Fry et al., 1985; Filmer and Kirschvink, 1989]. This coincidence leaves the possibility that the strata were remagnetized sometime during the late Cenozoic [Beck, 1991]. The presence of a widespread, 180-m thick intrabasinal olistostrome in middle Cretaceous rocks of the Valle Group on Cedros Island affords a rare opportunity to constrain the age of remanence aquisition to Late Cretaceous time using a slump test. An independent regional tilt test, using localities on different structural blocks on the island, also supports an early magnetization age.

We use the knowledge that shale compacts more than sandstone to test for inclination bias in the Valle Group. We would expect to see shallower average magnetic inclinations in shale than in sandstone if compaction were responsible for flattening the inclinations. We find no correlation between grain size and magnetic inclination in averages of 314 samples of the Valle Group on Cedros Island and the Vizcaino Peninsula. This result calls into doubt the suggestion of Butler et al. [1991] that compaction is responsible for the shallow inclinations in the Vizcaino terrane.

In summary, we find that neither remagnetization nor burial compaction can adequately explain shallow magnetic inclinations in the Valle Group. In the absence of other testable hypotheses explaining anomalously shallow magnetic inclinations in the Valle Group, the remanent magnetization measured on Cedros Island suggests that those strata were located about 20° farther south with respect to North America in middle Cretaceous time (Table 2). When Cedros Island sites are averaged with the combined data of Patterson [1984b] and Hagstrum et al. [1985] from the Vizcaino Peninsula, we calculate that the composite Vizcaino terrane was located $18 \pm 7°$ farther south with respect to North America in Cretaceous time (Table 2).

The tectonic paradox of Baja California still stands. Two tests of the paleomagnetic data reported here add strength to the far-traveled hypothesis for the Vizcaino terrane. We caution against applying these results to the entire Baja California Peninsula [e.g., Hagstrum and Filmer, 1990] until strong stratigraphic or petrologic links exist between the two regions. Jarrard [1986] showed that modern forearc regions commonly accommodate the strike-slip component of oblique convergence by moving as a crustal sliver broken from the plate edge. There presently is no irrefutable geologic link between the Vizcaino terrane and the rest of Baja California prior to deposition of late Cenozoic deposits

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TABLE 2. Paleomagnetic Summary of Vizcaino Terrane

Locality	N	I, deg	D, deg	a95, deg	k	Plat	Disp, deg
Cedros Island (beds)	42	34	..	5	13	19 ± 3	20 ± 6
Cedros Island (sites)	9	35	..	12	13	20 ± 8	20 ± 10
Vizcaino Peninsula (sites)	9	41	..	5	70	24 ± 5	16 ± 7
Vizcaino terrane	18	38	..	6	23	23 ± 5	18 ± 7
North America	..	58.7	341.5	4.9	352	39 ± 4	0

Plat is paleolatitude with 95% confidence interval; Disp is northward displacement with respect to North America using the methods of Beck [1989]; North America is expected results for Cedros Island (presently located at 28.2° N, 115.2° W.) in middle Cretaceous time, assuming contiguity with North America [Globerman and Irving, 1989]. Averages calculated with inclination values only [McFadden and Reid, 1982]. "Cedros Island (beds)" is average of 42 beds from Valle Group localities PO-1, VG-1, VG-2, and P-1; "Cedros Island (sites)" is average of nine sites from Valle Group. "Vizcaino Peninsula (sites)" is average of Valle Group sites from Patterson [1984b] (n=6) and nonremagnetized Valle Group sites from Hagstrum et al. [1985] (n=3); "Vizcaino terrane" is average of Cedros Island and Vizcaino Peninsula sites.

which cap both; the possibility of fairly recent relative motion between the Vizcaino terrane and the main body of Baja California still exists.

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